Albert Einstein’s brain: What have scientists discovered?

For decades, scientists have sought to understand the biological foundations of genius by examining the physical characteristics of Albert Einstein’s brain. Various studies indicate that while his brain weighed a normal amount, it possessed distinct anatomical features and enhanced connectivity that may have contributed to his extraordinary cognitive abilities. These findings offer a glimpse into the potential neural substrates of his mathematical and visuospatial prowess. Albert Einstein is widely recognized as one of the most influential physicists in history. In 1905, often referred to as his *annus mirabilis* or miraculous year, he published four groundbreaking papers that fundamentally altered the scientific understanding of the universe. These works addressed the photoelectric effect, Brownian motion, special relativity, and the equivalence of mass and energy. He later developed the general theory of relativity, cementing his reputation as a scientific revolutionary.

Because of his immense intellectual achievements, significant curiosity surrounded the source of his genius. Researchers and the public alike questioned whether his abilities stemmed from his environment and education or if he possessed a unique biological advantage. This debate regarding nature versus nurture prompted a desire to analyze his physical brain for clues. Einstein died on April 18, 1955, at Princeton Hospital due to a ruptured abdominal aortic aneurysm. During the autopsy, the pathologist Thomas Harvey removed Einstein’s brain for scientific study. Harvey weighed the organ, photographed it from multiple angles, and preserved it using formalin. He subsequently sectioned the brain into approximately 240 blocks and prepared histological slides, which are microscopic slices of tissue used for studying cellular structure.

In 1999, Sandra Witelson and her team at McMaster University published a study in *The Lancet* that focused on the gross anatomy of the brain. They compared Einstein’s brain with those of a control group. While the brain’s weight was not exceptional, the parietal lobes were about 15 percent wider than average.

Additionally, researchers discovered an unusual pattern in the Sylvian fissure. In most brains, this structure divides a region called the supramarginal gyrus. However, in Einstein’s brain, this division did not occur, potentially allowing for more efficient neural connections. This region is associated with visuospatial reasoning and mathematical thinking.

Later research in 2009 by Dean Falk identified a distinct knob-like structure in the motor cortex, an area responsible for voluntary movement. This feature is often observed in musicians and was likely linked to Einstein’s lifelong violin playing. The finding supports the idea of brain plasticity, meaning that the brain can change in response to repeated activity.

Further studies in 2013 examined the corpus callosum, which connects the two hemispheres of the brain. Researchers found that this structure was thicker than average, suggesting greater communication between the left and right hemispheres. This enhanced connectivity may have enabled Einstein to integrate analytical and creative thinking more effectively.

Questions 1–5: Multiple Choice

1. What is the main focus of the passage?
A. Einstein's life achievements
B. The debate about intelligence
C. Scientific studies on Einstein's brain
D. The history of neuroscience

2. What was normal about Einstein's brain?
A. Its shape
B. Its weight
C. Its structure
D. Its size

3. Why were scientists interested in Einstein's brain?
A. To compare it with animals
B. To understand the origin of genius
C. To preserve it for museums
D. To test medical equipment

4. What did Witelson's study reveal?
A. Einstein's brain was heavier
B. His motor cortex was larger
C. His parietal lobes were wider
D. His brain had fewer neurons

5. What is suggested about the corpus callosum?
A. It was smaller than average
B. It limited brain function
C. It enhanced communication between hemispheres
D. It was damaged

Questions 6–9: True / False / Not Given

6. Einstein’s brain was removed immediately after his death for research purposes.
7. All scientists agree that Einstein’s intelligence was purely genetic.
8. The Sylvian fissure in Einstein’s brain divided the supramarginal gyrus normally.
9. Einstein began playing the violin only in adulthood.

Questions 10–13: Short Answer (NO MORE THAN THREE WORDS)

10. In which year did Einstein publish four important papers?
11. What term is used to describe Einstein’s “miraculous year”?
12. Approximately how many blocks was Einstein’s brain divided into?
13. What structure connects the two hemispheres of the brain?